MIL-T-82394(TD)
15 December 1969
SUPERCEDING
NAVTRADEVCEN 3313-406
3 July 1961

MILITARY SPECIFICATION
TRAINING AID, TURBINE PRINCIPLE DEMONSTRATOR

This specification has been approved by the Naval Training Device Center, Department of the Navy.

1. SCOPE

1.1 This specification covers classroom training aid demonstrators for illustrating turbine principles of specific types.

1.2 Classification.-

1.2.1 Types.- This specification covers turbine principle demonstrators of the following types:

Type I, Velocity impulse turbine.

Type II, Reactor turbine.

Type III, Velocity and pressure compounded turbine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

PPP-B-601	Boxes Wood, Cleated-plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of

FSC 6910

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MIL-T-23991	Training Devices, Military; General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

943-208-011	Training Aid, Velocity Impulse, Turbine, Demonstrator, Type I (Device 29B6A)
943-208-021	Training Aid, Reaction Turbine, Demonstrator, Type II (Device 29B6B)
943-208-031	Training Aid, Velocity and Pressure Compound Turbine, Demonstrator, Type III (Device 29B6C)

PUBLICATIONS

Naval Training Device Center (NAVTRADEVCEN)

NAVSO-P-1381	Instructors Guide for Turbine Demonstrators, Devices 29B6A, B and C

PHOTOGRAPHS

Naval Training Device Center (NAVTRADEVCEN)

BXA-3437-11-57 BXA-3438-11-57 BXA-3439-11-57	Photographs, Views of Turbine Demonstrator Device 29B6A
BXA-3440-11-57 BXA-3441-11-57	Photographs, Views of Turbine Demonstrator Device 29B6B
BXA-3442-11-57 BXA-3443-11-57	Photographs, Views of Turbine Demonstrator Device 29B6C

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Unless otherwise specified herein materials shall be as specified on NAVTRADEVCEN drawings:

Type I - Drawing 943-208-011

Type II - Drawing 943-208-021

Type III - Drawing 943-208-031

Materials not specified on the drawings shall be in accordance with MIL-T-23991.

3.2 Design.- The design shall be in accordance with the drawings listed in 3.1. Functional description of each specific type of turbine demonstrator is outlined in NAVTRADEVCEN Publication NAVSO-P-1381. Typical views of assembly of the turbine demonstrators are shown on NAVTRADEVCEN Photographs:

Type I - Photographs	BXA 3437-11-57 through BXA 3439-11-57
Type II - Photographs	BXA 3440-11-57 through BXA 3441-11-57
Type III - Photographs	BXA 3442-11-57 and BXA 3443-11-57

3.2.1 Details of components.- Each training aid shall consist of the following major components:

(a) Transparent casing

(b) Rotor

(c) Blades, stationary and moving

(d) Bearing support

(e) Nozzle

(f) Steam chest

(g) Base

(h) Repair parts.

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3.2.2 Interchangeability.- All parts, subassemblies and assemblies having the same part number shall be directly and completely interchangeable with each other in respect to installation and performance, regardless of sources from which obtained without alteration and without selection for fit or performance.

3.3 Construction.- Construction shall be as shown on the drawings of 3.1. Where processes are not specified on the drawings, MIL-T-23991 shall apply.

3.3.1 Construction of the training aid shall permit removal of the upper half of each casing and removal of the rotor assembly. The rotor shaft and ball bearings shall be capable of removal as a unit from the case.

3.3.2 Prior to cementing, all contact surfaces of plastic pieces to be joined shall be machined or sanded. Whenever possible, a cushion surface shall be developed on one contact surface. Cementing solvents shall be used as recommended by the manufacturer for the particular plastic.

3.3.3 Rotor (moving) blades, stationary blades and nozzles shall be fabricated of molded or machined opaque plastic. Rotor blades shall be blue. Stationary blades and nozzles shall be red.

3.3.4 Casings, shrouds, rotor and rotor ends and steam chests shall be formed or fabricated of clear plastic.

3.3.5 Rotor assemblies shall be balanced and aligned.

3.3.6 Repair parts.- Each turbine demonstrator shall be provided with repair turbine blades and nozzles as follows:

(a) Type I turbine demonstrator:

(1) 10 stationary (red) blades

(2) 18 rotary (blue) blades

(b) Type II turbine demonstrator:

(1) 30 stationary (red) blades

(2) 30 rotary (blue) blades

(c) Type III turbine demonstrator:

(1) 10 stationary (red) blades

(2) 18 rotary (blue) blades

(3) 4 nozzles (red).

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3.4 Performance.-

3.4.1 The training aid shall operate on a minimum of 10 pounds per square inch (PSI) of air supply pressure to a maximum 20 PSI of air pressure. The source will be supplied by the training activity. A pressure relief valve shall be installed in the air supply line at the steam chest to limit the operating air pressure to 20 PSI maximum pressure.

3.4.2 Reliability.- The device shall withstand continuous classroom use and shall not be adversely affected in any manner during storage and transportation when under ambient conditions of temperature ranging from minus 65 to plus 165 degrees Fahrenheit, and of relative humidity ranging up to 95 percent. The device shall be constructed so that no components, assemblies or subassemblies shall bind or become undesirably free. No degradation shall be caused in the mechanical performance stipulated for the device because of shipment, handling or storage.

3.5 Dimensions and tolerances.- Dimensions and tolerances shall be as shown on the drawings of 3.1. Tolerances not shown on the drawings shall be such as to assure the interchangeability requirements of 3.2.2.

3.6 Finish and color.-

3.6.1 Finish.-

3.6.1.1 The base shall be sealed, primed and finished with a satin-surface lacquer applied in accordance with the requirements of MIL-T-23991.

3.6.1.2 Steam chests shall be ground or frosted finished in the areas indicated in the drawings of 3.1 to represent steam.

3.6.1.3 Outer periphery of rotors and outer rims of the turbine casings shall have a ground or frosted finish to make the plastic opaque in the areas indicated in the drawings of 3.1.

3.6.1.4 All aluminum supports and members shall be satin finish or anodized natural color.

3.6.2 Color.- Colors shall be as indicated on the drawings of 3.1.

3.7 Identification.-

3.7.1 Training device nameplate.- Unless otherwise specified by the procuring activity, an identifying nameplate shall be provided by the contractor in accordance with MIL-STD-130. The nameplate shall be affixed to the base in an accessible location.

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3.7.2 Markings of components.- All major components shall be labeled in accordance with the drawings of 3.1 and the photographs of 3.2.

3.7.2.1 Labels and lettering.- All labels shall be made of sheet or laminated plastic and shall be permanently attached to the appropriate component as indicated on the drawings. The information shall be engraved or silkscreened with black letters on white background or white letters on black background unless otherwise noted on the drawings.

3.8 Workmanship.- The training aid shall be constructed, assembled and finished with particular attention given to neatness, and thoroughness of machining of parts and assemblies, finishing and freedom of parts from burrs, sharp edges, tool marks and metal chips after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual, and mechanical examinations and testing of materials, subassemblies, parts, and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements specified herein.

4.4 Quality conformance inspection.-

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on the preproduction model and shall consist of all the examinations and tests specified herein.

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4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-approved preproduction model.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Dimensions and tolerances

(d) Finish and color

(e) Nameplate

(f) Workmanship

(g) Preparation for delivery.

4.6 Test.- The training aid shall be subjected to a continuous functional 15 minute operating test using varying air supply pressures from 10 PSI (minimum) to 20 PSI (maximum) to assure functioning and to demonstrate conformance with the requirements of 3.4.1. In addition the training aid shall be operated for a period of not less than 2 minutes under an air supply pressure of 35 PSI to prove mechanical safety of the unit and retested for 5 minutes using varying air supply pressures from 10 PSI (minimum) to 20 PSI (maximum) to assure performance integrity. No fixed part or assembly shall become loose, sluggish or show degradation during the test. Rotor assemblies shall maintain balance and alignment. Relief valves shall operate within the required limits. Failure of any component shall be cause of rejection and requires the test to be repeated after the necessary repair, replacement or adjustment has been made.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C (see 6.2).

5.1.1 Level A.- For level A, each trainer shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method 1A8.

5.1.1.1 Unit packaging.- Each trainer shall be packaged one unit to a container conforming to PPP-B-636.

5.1.1.1.1 Repair parts as specified in 3.3.6, to be supplied with each trainer shall be packaged in separate contrainers conforming to PPP-B-636 and packed with the unit.

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5.1.2 Level C.- For level C, preservation and packaging shall conform to manufacturer's commercial practice.

5.2 Packing.- Packing shall be level A or C (see 6.2).

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621 or PPP-B-601, Styles A or B with steel banding.

5.2.2 Level C.- Units packaged in accordance with 5.1.2 shall be packed in containers which will ensure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment, handling and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid will be used to provide student instruction in the theory and principles of operation of three (3) specific types of steam turbines.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Procuring activity designation (device number or other) (see 6.4)

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2)

(d) Nameplate data.

6.3 Definitions.- Except as specified in 6.3.1 and 6.3.2 definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction model.- A model suitable for complete evaluation of mechanical form, design and performance. It shall be of mechanical form, employ standard or nonstandard parts (approved by the agency concerned) and be completely representative of final equipment.

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6.3.2 Production unit.- A unit in its final mechanical form of final production design made by production tools, jigs, fixtures and methods.

6.4 Naval Training Device Center Designations.- The types of turbine demonstrators specified in 1.2.1 are designated by the Naval Training Device Center as follows:

Type I - Device 29B6A
Type II - Device 29B6B
Type III - Device 29B6C

CUSTODIAN
Navy-TD

PREPARING ACTIVITY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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